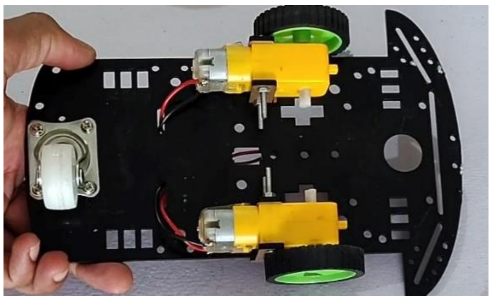
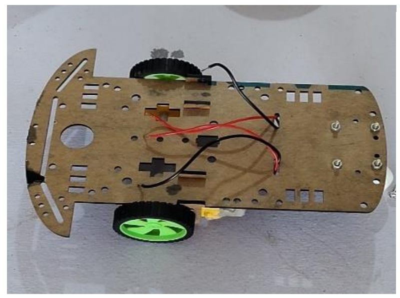
Circuitly.in SmartTrack Installation KIT Steps

1 . Installation of motor and wheels to chassis

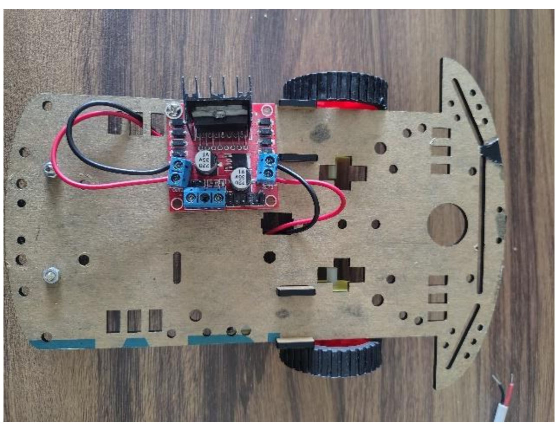




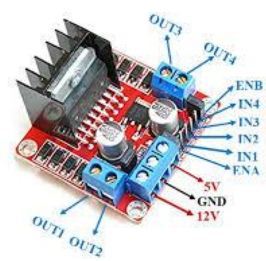
Motor Holder

- To install a motor on chassis first using motor holder and long nut and bolts
- Caster wheel nut should be tightened on top side of chassis so that wheels can freely move.

2 . Installation of Motor Driver and its connection with motor

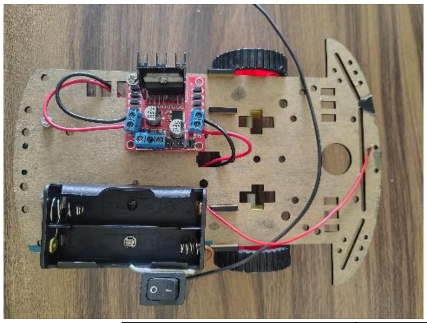


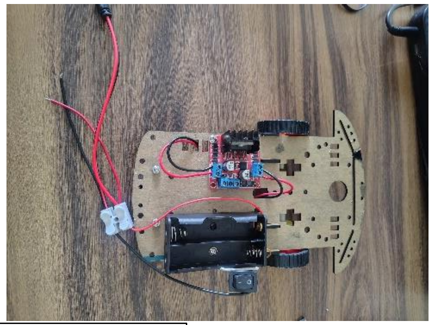
Jumper PIN

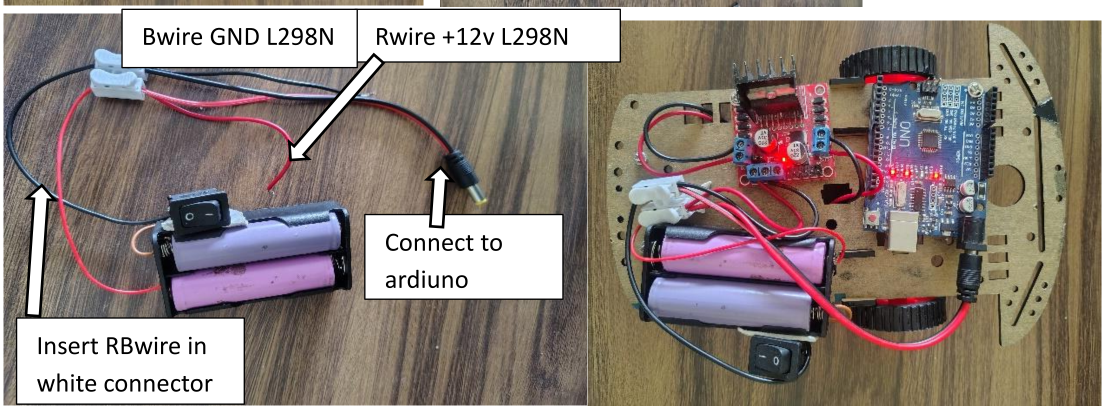


- Place motor controller L298N on Chassis & fixed it position using nut or bolt.
- Insert the Left motor wire out1, out2 blue connector and tightened it using screw driver.
- Insert the Right motor wire out3, out4 blue connector and tightened it using screwdriver.
- Remove the jumper Pin of ENA and ENB on L298N driver (Check image L298n)

3 . Installation of Battery Holder & its connection with L298N and ardiuno

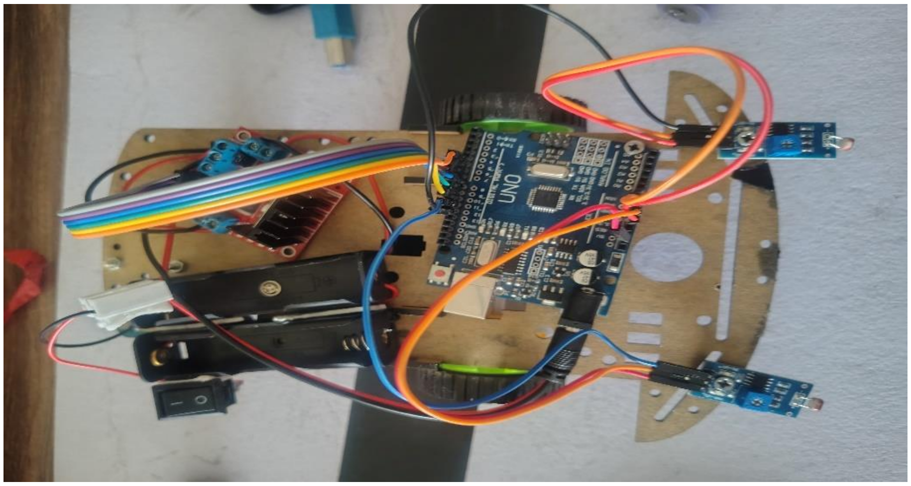






- Fix the position on battery holder using nut and bolt.
- Form battery holder 2 output red,black wire are coming. Insert this wire white connecter
- At end of white connecter it has 3 wire separate red,black audio jack.
- Separate red wire connect to 12v of L298n and black wire connect it to GND of L298n
- Fixed the position of Arduino on chassis and powerup Arduino using audio jack from battery.
- When switched on LED on motor driver L298n and Arduino will glow.
- **Note Above are common connection for all robot don't change it.**
- **Reverse motor connection if motor rotate in opposite direction.**

4. connection of Light follower robot

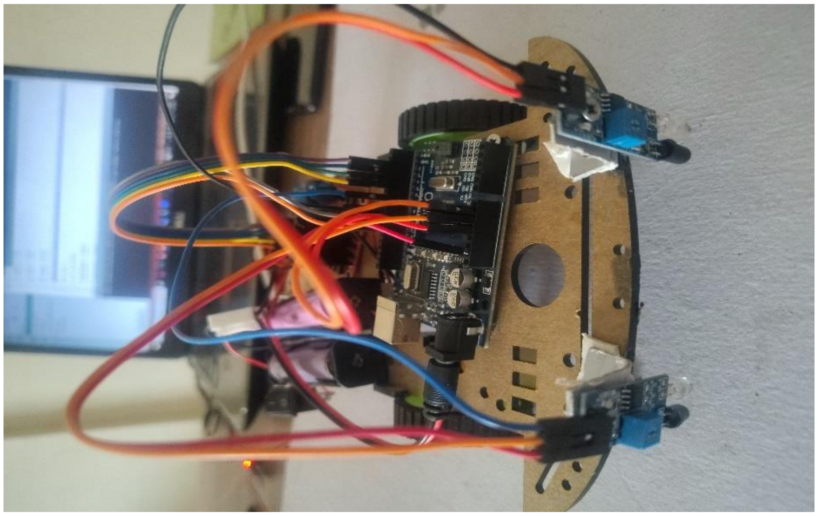


- Secure the position of LDR sensor on chassis and do the connection as mention in the table with Arduino & L298N motor using Jumper wires.

- Pin connection with Arduino
-

L298N driver Pins	**Arduino Pins**
ENA	10
IN1	2
IN2	3
IN3	4
IN4	5
ENB	11
LDR Sensor	**Arduino Pins**
VCC	5V
GND	GND
Do(Right LDR)	7
DO(Left LDR)	8

5 Connection with IR Sensor

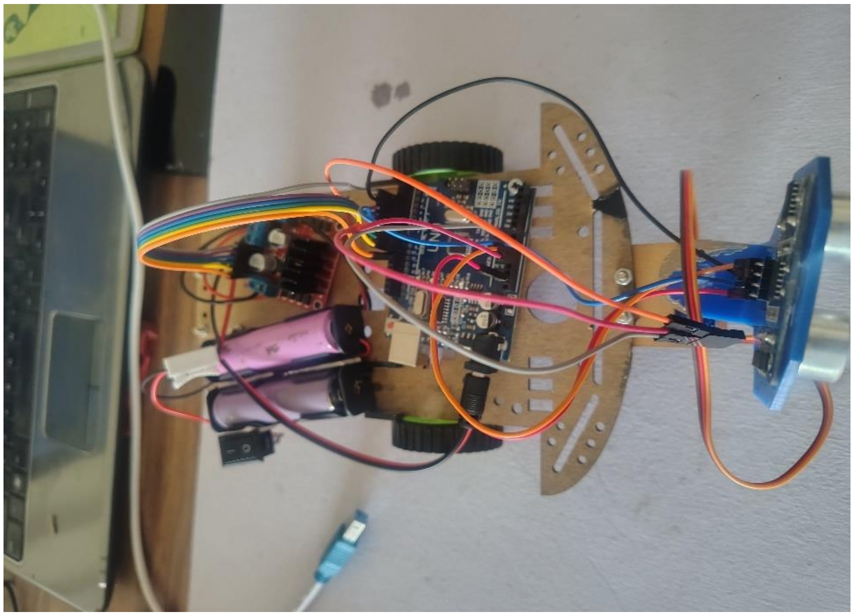


- Secure the position of IR sensor on chassis and do the connection as mention in the table with Arduino & L298N motor using Jumper wires.

- Pin connection with Arduino

L298N driver Pins	Arduino Pins
ENA	5
IN1	9
IN2	10
IN3	7
IN4	8
ENB	6
IR Sensor	**Arduino Pins**
VCC	5V
GND	GND
Do(Right IR)	11
DO(Left IR)	12

5. Connection for Obstacle avoidance robot



Servo motor mount on front of chassis

- Secure the position of ultronics sensor and SG90 servo on chassis and do the connection as mention in the table with Arduino & L298N motor using Jumper wires.

Connection Pin out

L298N driver Pins	**Arduino Pins**
ENA	5
IN1	2
IN2	3
IN3	7
IN4	8
ENB	6
Ultrasonic Sensor	**Arduino Pins**
VCC	5V
GND	GND
TRIG	A0
ECHO	A1
SG90	**Arduino Pins**
VCC	5V
GND	GND
Signal	10

To charge battery use the charging module with cell holder

- **Use only 5V charger , if while charging cell getting hot stop charging .**